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Project Work Proposal Defense

On

Mero Diary: A Secure Web-Based Personal Journaling and Activity Recommendation Application

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B.Sc. CSIT 4th year/ 7th Semester(2079B.S)

Magh 2026

Presentation Outline

- About Project Topic
- Literature Review
- Methodology
- Time Scheduling
- Conclusion
- References and Bibliography

About Project Topic

- Mero Diary is a secure web-based personal journaling system.
- Allows users to record daily life events, activities, and health-related data.
- Focuses strongly on privacy, confidentiality, and data security.
- Uses encryption and secure authentication mechanisms.
- Provides personalized activity recommendations to support user well-being.

Problem Statement

- Users hesitate to store personal thoughts online due to privacy and security concerns.
- Existing diary apps lack meaningful analysis and wellness guidance.
- Manual journaling does not help users improve habits effectively.

Objectives

- Provide a secure platform for digital journaling.
- Protect user data using encryption and secure authentication.
- Generate personalized activity recommendations for better well-being.

Scope

- Secure login, encrypted diary entries.
- Daily journaling and wellness tracking.
- Basic data visualization and recommendations.
- Recommendation of activities

Limitations

- Recommendation accuracy improves only with sufficient user dataset
- Not feasible for more than 10 users due to resource constraint

Review of Existing Systems

Day One Journal

- A popular digital journaling application used across multiple platforms.
- Provides strong encryption to protect user diary entries.
- Includes mood tracking features and guided prompts for daily reflection.
- However, it does not offer predictive or data-driven wellness recommendations based on user behavior or historical data.

Penzu

- A secure online diary platform focused primarily on privacy.
- Uses AES-256 zero-knowledge encryption, ensuring that even the service provider cannot access user content.
- While it ensures high confidentiality, it lacks analytics and personalized insights to help users improve habits or well-being.

Study of current system

- **Security Focus in Digital Journals:** Existing systems prioritize encryption and authentication (AES, bcrypt, JWT), but often overlook personalized wellness analytics and predictive activity suggestions [1], [
- **AES-128 Effectiveness:** AES-128 combined with HTTPS and client-side key management provides secure, real-time journal entry protection with low processing time (~0.84 seconds), suitable for resource-constrained environments [1], [5].
- **Limitations of Commercial Apps:** Apps like Day One and Penzu offer end-to-end encryption and mood tracking, but lack predictive activity suggestions and analytics-driven well-being insights [2], [4].
- **Advanced Authentication Methods:** JWT authentication with bcrypt hashing enables secure, stateless, cross-device user access, outperforming traditional session-based models [3], [8].
- **Predictive Analytics:** Linear regression can quantify behavioral data to predict wellness outcomes, forming a basis for activity optimization [6].
- **Hybrid Recommendation Systems:** Combining rule-based logic, regression, and ensemble methods
- like random forests improves personalized wellness recommendations compared to single-model
- approaches [7].

Security Techniques from Literature

- AES encryption is widely recognized as an efficient and reliable method for securing sensitive data at rest.
- HTTPS is essential for protecting data during transmission and preventing unauthorized access.
- Bcrypt password hashing and JWT authentication provide scalable, stateless, and secure user authentication mechanisms suitable for modern web applications.

Research Gaps Identified

•Most journaling apps focus only on storage and privacy.
•Very limited use of analytics for well-being improvement.

Proposed Improvement

- Hybrid recommendation system using:
  - Rule-based logic
  - Linear regression
  - Random Forest algorithms
- Combines security with wellness insights.

Methodology

Requirement Elicitation

- Literature review of existing journaling apps.
- Study of previous research papers.
- Observation of user journaling behavior.

Requirements (Hardware/Software)

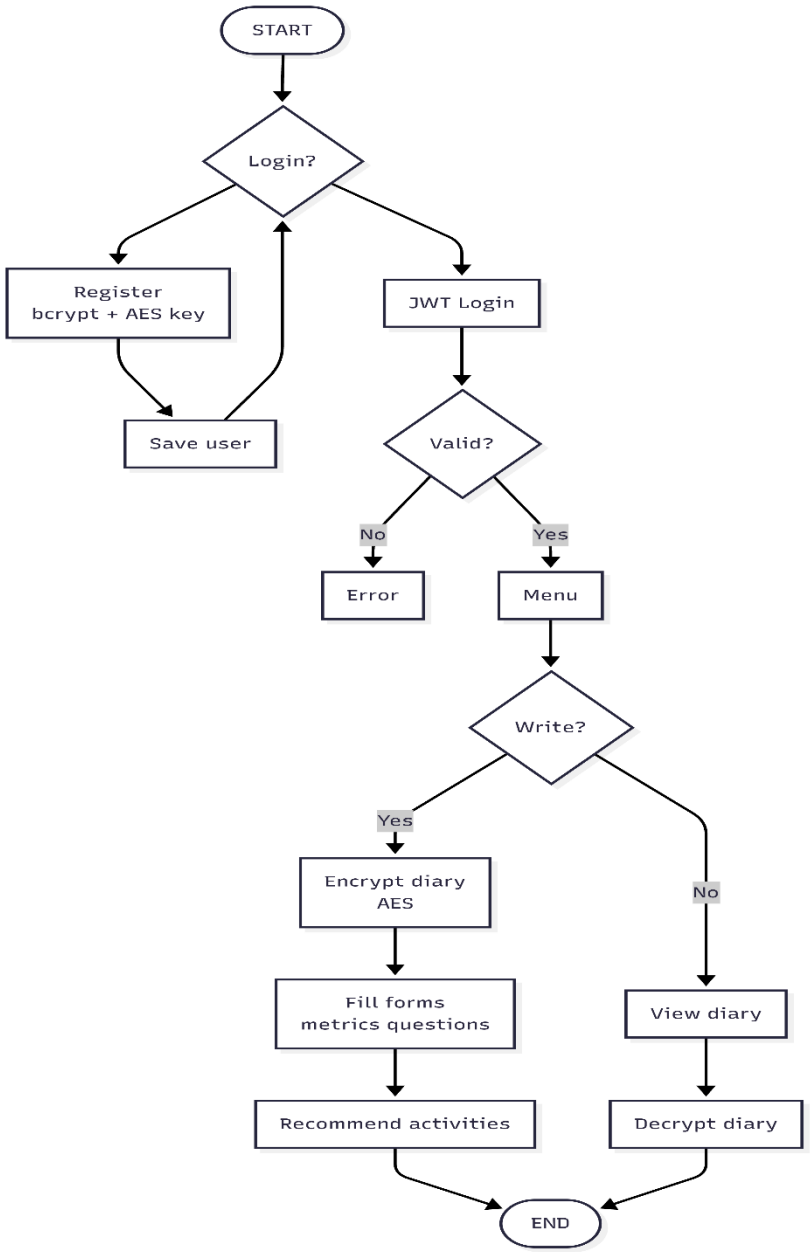
Hardware

- Laptop/Desktop
- Internet connection

Software

- Frontend: ReactJS
- Backend: Node.js, Express
- Database: Firebase/Supabase
- Security: bcrypt, JWT, AES
- Recommendation Engine: Python

flowchart


START
Login?
Register
bcrypt + AES key
Save user
JWT Login
Valid?
No
Yes
Error
Menu
Write?
Yes
No
Encrypt diary
AES
Fill forms
metrics questions
Recommend activities
View diary
Decrypt diary
END

Process Model

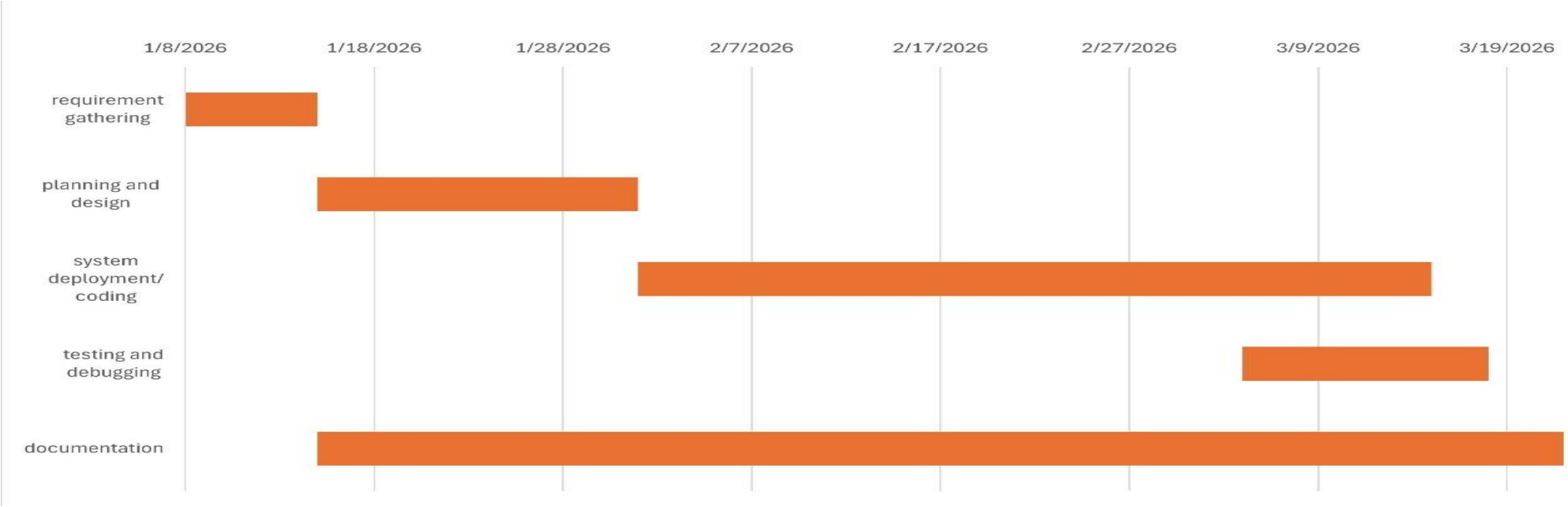
Agile Model

- Easy iteration
- Continuous testing and improvement

Time Scheduling

Gantt Chart

- Requirement Analysis – Week 1
- Design – Week 2
- Development – Week 3–4
- Testing – Week 5
- Final Presentation – Week 6



Conclusion

- Mero Diary ensures secure and private digital journaling.
- Combines encryption with wellness analytics.
- Encourages healthier habits through recommendations.

Expected Outcomes

- Secure web-based diary system.
- Encrypted user data and safe authentication.
- Personalized wellness activity suggestions.
- Visual insights into user habits.

References

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